UNITED STATES OF AMERICA
BEFORE THE
FEDERAL ENERGY REGULATORY COMMISSION

Interconnection of Large Loads to the Interstate Transmission System	)))	**Docket No. RM26-4-000**

COMMENTS OF
THE CALIFORNIA INDEPENDENT SYSTEM OPERATOR CORPORATION
ON THE ADVANCED NOTICE OF PROPOSED RULEMAKING

The California Independent System Operator Corporation ("CAISO") submits these comments in response to the Federal Energy Regulatory Commission's ("Commission") Notice Inviting Comments, issued on October 27, 2025, regarding the Advanced Notice of Proposed Rulemaking ("ANOPR") released on October 23, 2025 in the captioned docket.[1]

I. EXECUTIVE SUMMARY

The CAISO shares the ANOPR's stated goals of ensuring the timely and orderly interconnection of large loads while respecting States' exclusive jurisdiction over retail electricity sales, the siting and expansion of generation facilities, and distribution infrastructure. The CAISO acknowledges the jurisdictional concerns raised in the ANOPR itself and by States in response,[2] and understands that large loads present a unique set of challenges because, while they are retail customers, they interconnect directly to the transmission system and can impact reliability in ways comparable to

1 Capitalized terms not otherwise defined herein have the meaning set forth in the CAISO tariff, and references to specific sections, articles, and appendices are references to sections, articles, and appendices in the current CAISO tariff unless otherwise indicated.

2 https://pubs.naruc.org/pub/2C526A94-D533-BE0A-336A-178A366C7A91.

large wholesale generators. The CAISO does not opine on the jurisdictional legal issues large loads present, but is confident that they will be resolved consistent with the Federal Power Act, including its respect for States' rights. The CAISO offers these comments to the Commission, States, NERC, and other policymakers to explain the challenges large loads present to grid and market operators like the CAISO, and the reforms the CAISO believes are necessary to ensure reliability, affordability, and efficient markets. As the home of Silicon Valley, California has led the way in facilitating technology and data centers. The CAISO is eager to work with policymakers to continue these efforts. The CAISO believes collaboration among the Commission, States, transmission providers, and industry stakeholders will remain essential to facilitate the timely interconnection of large loads. To this end, the CAISO generally agrees with several of the substantive principles outlined in the ANOPR, and views them as measured and equitable. Reforms grounded in the existing rules and procedures applicable to large, transmission-connected generators can be a pragmatic approach that can benefit the broader industry by providing consistency and clarity in the interconnection process for large loads.

II. COMMENTS

A. Large Loads' Impact on Reliability and the Wholesale Markets

As an independent system operator, balancing authority, and reliability coordinator, the CAISO's foremost responsibility is maintaining reliability across the Western interconnection. Large loads directly interconnected to the transmission system have a direct and substantial impact on that responsibility.

Large loads present new challenges for long-term planning, short-term forecasting, and grid stability. NERC has already described these challenges in detail,[3] finding, *inter alia*:

- "Many emerging large loads differ technologically from past industrial loads, leading to significant uncertainty in modeling their behavior and consumption profiles."[4]
- "[H]igh-intensity electrical processing loads . . . often experience frequent on-off cycles, causing large, random fluctuations in consumption. Such behavior can lead to dispatch errors, shutting down units prematurely, committing units that are not needed, and generally causing balancing issues."[5]
- "Large rapid changes in demand will also rapidly alter the flow of reactive power in the system, potentially exceeding the ability of existing voltage controls to respond and lead to over- and undervoltage conditions. This could lead to the unplanned outages of generation plants initiated by voltage-ride through protection, load loss related to undervoltage load shedding, or even more widespread outages caused by voltage instability and collapse."[6]
- "If enough large load energizes and is not tripped off-line, the frequency may become low enough that generators trip off-line due to underfrequency protection. If this occurs, frequency instability and widespread outages are a

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[3] NERC, "Characterization and Risks of Emerging Large Loads," at 15 *et seq.*, July 2025, https://www.nerc.com/comm/RSTCReviewItems/3_Doc_White%20Paper%20Characteristics%20and%20Risks%20of%20Emerging%20Large%20Loads.pdf.

[4] *Id*. at 18.

[5] *Id*.

[6] *Id*. at 20.

serious possibility as each generator that trips off-line further reduces the system's ability to halt the decline in frequency."[7]

NERC is raising legitimate concerns. There have already been instances of large loads' impact on reliability in the Eastern Interconnection.[8] Because grid operators must keep supply and demand perfectly balanced at all times, large loads present the same issues to reliability as large generators.

Large loads' direct impact on wholesale markets likewise is considerable. Sudden or sustained demand spikes from data centers directly impact transmission congestion, resulting in more volatile pricing. Grid operators must procure additional reserves, fast-ramping resources, and frequency response to address the contingencies of demand fluctuations or sudden unplanned outages from large loads. And large loads will require more generating capacity and transmission capacity to meet demand.

B. Principles for Reform

1. Definition

The ANOPR proposes that Commission reform would be limited to interconnections directly to the transmission system, using the seven-factor test.[9] For any reform, the CAISO supports the use of clear, well-defined lines the industry already uses and understands. As the Commission, States, and transmission providers work to interconnect large loads, they should use easily understandable rules rather than

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7 *Id*. at 25.

8 *See, e.g.*, NERC, "Incident Review: Considering Simultaneous Voltage-Sensitive Load Reductions," https://www.nerc.com/pa/rrm/ea/Documents/Incident_Review_Large_Load_Loss.pdf#search=load%20loss%20data%20center.

9 ANOPR at P 18.

complex standards to reduce regulatory uncertainty. The CAISO agrees that large loads on the transmission system should be a priority: Because the largest loads will require the voltages only the transmission system can provide, they present greater challenges than loads that will connect to distribution systems. The CAISO supports the ANOPR principle coupled with the Commission's other qualifications in the ANOPR and its recent decision in *Tri-State Generation and Transmission Association*, namely, that "a regulation that 'specifies terms of sale at retail' would exceed the Commission's authority under the FPA," regardless of the level of interconnection.

2. 20 MW Threshold

The ANOPR proposes that its reforms should only apply to new loads greater than 20 MW and, for hybrid facilities, where the load is greater than 20 MW.[10] The CAISO appreciates this clear demarcation: Any final rule should have a clear, easily understandable rule for identifying large loads, tailored to the specific challenges the Rule is intended to address. Although a broad standard like the definition in the NERC white paper avoids false positives and false negatives in theory,[11] such standards are difficult to apply in practice and often lead to disputes. The 20 MW threshold has proven to be a reliable figure to assess the risk a generation or storage facility represents to the transmission grid. For grid operators and planners, loads present the same challenges as generators in maintaining the balance between supply and

[10] ANOPR at P 19.

[11] Only for purposes of its discussion in the white paper, NERC used the large load definition, "Any commercial or industrial individual load facility or aggregation of load facilities at a single site behind one or more point(s) of interconnection that can pose reliability risks to the BPS due to its demand, operational characteristics, or other factors." NERC would likely agree that such a definition would be impractical in a final rule, reliability standard, or tariff, requiring a "I know it when I see it" approach.

demand, so sharing the same threshold is sensible. Although any demarcation will affect development to some extent, much higher MW thresholds may incentivize developers to create multiple facilities each right under the limit to avoid jurisdiction, rather than a single facility that benefits the facility and the grid from economy of scale. The 20 MW threshold avoids this incentive without creating barriers to smaller facilities.

3. Coordinated Studies

The ANOPR proposes that jurisdictional utilities should study load and hybrid facilities together with generating facilities.[12] Today, transmission providers must account for co-located retail loads subject to state-jurisdictional tariffs but using FERC-jurisdictional tariffs for generator interconnections. For example, the CAISO tariff currently does not have a mechanism to consider large loads in the generator interconnection study process. Providing a single study process for large load, hybrid, and generating facilities may simplify many aspects for transmission providers and developers. At the same time, it could introduce new complexities to the interconnection study process and transmission planning process, both of which have been significantly reformed by recent Commission orders. The longstanding approach of enabling transmission providers to propose tailored processes to meet regional needs under the independent entity standard will allow transmission providers to tailor their studies to ensure the orderly and timely interconnection of large loads.

The CAISO works closely with utilities and State energy agencies to incorporate large loads into its transmission planning process, which is integrated into the generator

[12] ANOPR at P 20.

interconnection process. As set out in a 2022 Memorandum of Understanding, the California Energy Commission (CEC) develops a long-term statewide energy demand forecast, which incorporates information from utilities across the ISO footprint.[13] The CEC's demand forecast includes large loads mapped to substation locations used in the CAISO's transmission studies. Using the CEC's forecast, the California Public Utilities Commission (CPUC) provides resource planning information to the CAISO for its annual transmission planning process. The CAISO also receives resource planning information from publicly owned utilities. The CAISO uses these inputs to develop a final transmission plan and initiate development of the recommended transmission projects. Developers and stakeholders then use the transmission plan to understand where new transmission will be constructed and what capacity will be available. The CAISO explained these processes in detail in a letter responding to Commissioner Rosner's inquiry to ISO/RTOs. Currently, the CAISO, CPUC, and CEC forecasts and planning information indicate that there is and will be adequate generating capacity to meet increasing demand from large load growth.

4. Study Requirements

The ANOPR proposes that, like generating facilities, load and hybrid facilities should be subject to standardized study deposits, readiness requirements, and withdrawal penalties.[14] The CAISO supports this proposal, and recommends that if the Commission ultimately mandates specific reforms on this area, that any final rule enable

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[13] https://www.caiso.com/documents/iso-cec-and-cpuc-memorandum-of-understanding-dec-2022.pdf.

[14] ANOPR at P 21.

transmission providers to align the large load processes with the processes transmission providers have established in complying with Order No. 2023. Having the flexibility to align any large load processes with reforms already accepted and implemented by transmission providers in close coordination with their State policymakers will further the goals of the ANOPR. Large load interconnection studies may require specific mechanisms for transmission owners experienced in conducting such studies, as well as mechanisms to avoid the State-jurisdictional concerns that led to the Commission's rejecting the rules proposed in *Tri-State.* This flexibility will facilitate a faster compliance process to help large loads interconnect as soon as possible and with greater regulatory certainty.

5. Study Assumptions

The ANOPR proposes that hybrid facilities should be studied based on the amount of injection and/or withdrawal rights requested.[15] For example, a hybrid facility consisting of a 500 MW load and a 600 MW generating facility may seek no withdrawal rights and 100 MW of injection rights. The CAISO generally supports this proposal so long as it remains strongly coupled with principle six to maintain reliability. Hybrid facilities that have strict, unwavering protections like circuit breakers at their point of interconnection should be allowed to have network upgrades based on their net rights requested. This will save ratepayers and developers from unnecessary costs and help to avoid long construction timelines. Without such protection facilities, however, it would not be possible to interconnect hybrid facilities safely and reliably. The generator or

[15] ANOPR at P 22.

load could go offline, causing more demand or energy to hit the grid than what the interconnection studies contemplated. This would have significant safety and reliability implications no financial penalty could later remedy.

In other words, the ANOPR's use of the term "study" is problematic. The CAISO understands the intent of this principle to mean that hybrid facilities with strict injection/withdrawal protections should be able to interconnect with only those network upgrades necessary based on the requested net rights. This is similar to how large generators can request less interconnection capacity than actual generating capacity so long as they install protection facilities at this point of interconnection. Any final rule should clarify that transmission providers still must have detailed modeling information to "study" the full load and generating capabilities of each individual part of the facility. Conducting short-circuit duty contribution and dynamic stability studies, for example, is not possible without detailed models of the individual generators and loads at the site. In order to assign the correct protection facilities consistent with principle six, the transmission provider must study the gross, not net, components of the facility. Having more data in the interconnection study inputs will enable faster and more reliable interconnections for large loads.

6. Protection Facilities

The ANOPR proposes that any hybrid interconnection shall be required to install the system protection facilities necessary to prevent unauthorized injections or withdrawals that exceed the respective rights.[16] The ANOPR also seeks comment on

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16 ANOPR at P 23.

whether other operational limitations should be considered, and the minimum technical requirements for such system protection facilities.[17] The CAISO believes this principle is among the most critical to establish. As NERC has already demonstrated, large loads can affect frequency, voltage, and power quality just as large generators do.[18] The CAISO thus recommends the applicable operational requirements from Section 9 of the *pro forma* Large Generator Interconnection Procedures on large loads, as an example of useful rules already vetted by the industry. This will ensure that large loads do not affect reliability, nor pass undue costs to generators, utilities, and other ratepayers to make up the difference from large loads' impact on the grid.

The CAISO also recommends that any reform include the requirement that large loads provide load profiles and expected behavior models as part of the interconnection process. Many large loads, such as artificial intelligence training centers, operate in novel ways that impact the grid. Transmission providers and stakeholders will need operational data to understand how these large loads can impact reliability. To support reliable operations and planning, large loads should be required to submit detailed steady-state, dynamic (RMS), and, for certain facilities, electromagnetic transient (EMT) models reflecting expected demand behavior, similar to the modeling requirements currently applied to inverter-based generators. This includes representation of load ramps, start-up sequences, step-change behavior, and harmonic characteristics.

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[17] *Id*.

[18] NERC, "Characterization and Risks of Emerging Large Loads," at 15 *et seq.*, July 2025, https://www.nerc.com/comm/RSTCReviewItems/3_Doc_White%20Paper%20Characteristics%20and%20Risks%20of%20Emerging%20Large%20Loads.pdf.

Similarly, large loads should be required to provide transmission providers with real-time direct telemetry and outage schedules. Transmission providers like the CAISO currently have limited visibility into the operations of any specific end user. Each large load can have a singular impact on reliability and markets because of their capacity and direct interconnection to the transmission grid. Transmission providers thus need a similar level of visibility of large loads as they have for large generators. The CAISO recommends that any final rule require large loads and hybrid facilities to provide transmission providers with real-time telemetry at the component level. This would enable transmission providers to monitor performance of the large load and any generating units on the facility, just as transmission providers do today for large generators. Given the scale of these facilities, providing telemetry would be a minor cost with significant benefits to reliability. The CAISO does not believe that large loads need scheduling coordinators unless the large loads or co-located generators intend to be dispatchable or participate in the wholesale markets.

The ANOPR also seeks comment on whether a hybrid interconnection customer should be subject to penalties for unauthorized injections or withdrawals, how any such penalties should be designed, and how such penalties should be allocated to other transmission customers.[19] Again, the CAISO believes that large loads and hybrid facilities should be under the same rules as large generators. The CAISO's experience with large generators demonstrates that specific penalties for unauthorized injections or withdrawals are unnecessary. System protection facilities should physically prevent injections or withdrawals that substantially impact the transmission system, including

[19] *Id*.

circuit breakers that would isolate the facility from the grid before it could affect reliability. For other unauthorized behavior, transmission providers can report the facility to the Commission's Office of Enforcement and other regulatory enforcement agencies.

7. Curtailable and Hybrid Timelines

The ANOPR proposes that the interconnection study of large loads that agree to be curtailable and hybrid facilities that agree to be curtailable and dispatchable should be expedited.[20] The CAISO agrees that it is generally easier to study and interconnect curtailable facilities compared to firm loads and non-curtailable facilities; however the details of how such resources would or could be expedited deserves careful consideration to prevent unintended delays to other projects and processes. The ability to curtail and dispatch these facilities gives transmission providers many tools to maintain reliability that firm and non-curtailable facilities lack, especially the ability to curtail the facility in the event of a grid contingency. An expedited process would appropriately incentivize more developers to consider curtailable and dispatchable facilities, which ultimately will benefit large loads, the grid, and ratepayers. Although the specific grid topology will always be a key factor, firm and non-curtailable facilities will usually require more network upgrades and system protections to maintain reliability.

The CAISO recommends that any final rule should establish distinct "lanes" based on the level of dispatch tools the transmission provider will have—all, some, or none. For example, for the most "expedited" interconnection, large loads would have to

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20 ANOPR at P 24.

be completely curtailable or dispatchable. In other words, being curtailable on some hours or on some days or for only a portion of the load or generating capacity should not allow a facility to avail itself of the same expedited process as completely curtailable facilities.

The ANOPR also seeks comment on whether this curtailable large load studies should be accomplished through a serial interconnection study process or by some other means; and whether such studies can be completed in 60 days. The CAISO defers to its transmission owners on these questions because they have extensive experience with large load studies.

8. Cost Responsibility

The ANOPR proposes that load and hybrid facilities should be responsible for 100% of the network upgrades that they are assigned through the interconnection studies.[21] The ANOPR also seeks comment on whether such costs should be offset through a crediting mechanism and, if so, over how many years. The CAISO generally defers to States and their experience with load ratemaking on this issue; however, the CAISO requests that any final rule provide clear guidance on cost responsibility for hybrid facilities: Separating which network upgrades were required by a generator and which network upgrades were required by a co-located large load is impractical, if at all possible for some types of upgrades. Any final rule should at a minimum establish which cost responsibility rules apply to hybrid facilities: those for generators or those for large loads. The CAISO also recommends that the Commission and States clarify how

[21] ANOPR at P 25.

any final rule applies for generating facilities engaged in wholesale sales co-located with large loads compared to generating facilities that only serve the co-located large loads.

9. Equal Access

The ANOPR proposes that to the extent the interconnection customer is not the transmission owner, the interconnection customer shall be afforded the same (or equivalent) option to build as currently provided to generator interconnection customers.[22] The CAISO supports this proposal, as equal access is a foundational requirement for transmission providers. The CAISO also notes that in situations where the CAISO transmission planning process identifies new network upgrades to support federal, State, or local public policies to enable large loads, those network upgrades would be subject to the same competitive solicitation rules as other public policy network upgrades.[23]

10. Partial Suspensions to Accommodate Large Loads

The ANOPR proposes that an existing generating facility that seeks to enter a partial suspension to serve a new load at the same location must go through a system support resource (SSR)/reliability must run (RMR) type study.[24] The ANOPR also invites comments on whether and how resource adequacy should be considered in the SSR/RMR type study. The CAISO supports this proposal. The CAISO notes that its

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22 ANOPR at P 26.

23 *See* Section 24.5.1 of the CAISO tariff.

24 ANOPR at P 27. ("The study must consider system conditions, including forecasted load growth, at least three years after the proposed suspension date. The partial suspension can only proceed after any network upgrades needed to ensure reliability are placed into service. Any such network upgrades shall be the responsibility of the generating facility.")

reliability must run studies account for local and system capacity to ensure reliability. Local capacity needs are among the common reasons for designating a reliability must run resource, so they should be included in any study for a generator electing to change its participation model. Excluding capacity issues, especially on a local reliability level, could jeopardize reliability in the short-term until replacement generation comes online.

11. Transmission Service

The ANOPR proposes that utilities serving large loads, including those at hybrid facilities, should be responsible for transmission service based on their withdrawal rights, as that value amount reflects the quantity of capacity and energy that is being transmitted across the transmission system to the load.[25] The CAISO supports this proposal and encourages providing flexibility to transmission providers and utilities to develop planning processes appropriately. Correctly incentivizing large load developers to right-size their withdrawal and injection rights will help States, transmission planners, and load-serving entities avoid unnecessary network upgrades and additional generation procurement at ratepayers' expense. Incentivizing hybrid facilities to inject more energy or curtail demand at key times, especially for contingencies, will do the same.

12. Ancillary Services

The ANOPR proposes that utilities serving large loads, including those at hybrid facilities, should be responsible for ancillary services based on peak demand, without

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[25] ANOPR at P 28.

consideration of any co-located generation.[26] The ANOPR states that any co-located generating facilities will similarly be fully compensated for the provision of ancillary services. The CAISO supports this proposal as all load using the transmission grid should be responsible for procuring ancillary services. At the same time, the ANOPR appropriately incentivizes co-located generation to provide ancillary services when able to do so.

13. Transition Plan

The ANOPR proposes that there must be a plan to implement these proposed reforms.[27] The ANOPR also seeks comment on appropriate transition plans, including the treatment of large load interconnections that are already being studied for interconnection. The CAISO supports this proposal, and recommends an approach similar to the transition mechanisms used in Order No. 2023. Order No. 2023 provided *pro forma* transitional mechanisms and timelines, but still afforded transmission providers broad discretion to transition to the new rules based on their individual circumstances.[28] Similar flexibility will enable transmission providers and developers to ensure the timely and orderly interconnection of large loads.

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[26] ANOPR at P 29.

[27] ANOPR at P 30.

[28] For example, the Commission approved the CAISO and New York ISO's (NYISO) elections to forego transitional groups and extra time afforded to transition. *See California Independent System Operator Corp.*, 191 FERC ¶ 61,119 at P 143 (2025); *New York Independent System Operator Inc.*, 191 FERC ¶ 61,049 at P 397 (2025). The Commission accepted the CAISO and NYISO proposed effective dates for their new procedures on the days immediately after their respective filings.

14. NERC Requirements

The ANOPR proposes that utilities serving large loads must meet all applicable NERC reliability standards and OATT provisions.[29] The ANOPR states that "Utilities and we must be prepared to revise large load interconnection procedures and agreements, as necessary."[30] It goes on to state that NERC should review its reliability standards to determine if new registration categories or new or modified reliability standards are required to ensure reliability of the BES.

The CAISO supports this principle. As discussed above, NERC and others have already observed that large loads present many novel challenges for maintaining reliability. Given the rapid proliferation of these large loads, it is critical for the Commission, NERC, States, and utilities to adapt their standards and procedures as necessary to maintain reliability. The CAISO supports the efforts of NERC's large load taskforce, including the recent NERC Alert seeking data on large loads and utility procedures.[31] Keeping reliability standards reform aligned with the reforms in this proceeding—both substantively and temporally—will help transmission providers implement those reforms as soon as possible. It also will provide developers regulatory certainty, and will facilitate the timely interconnections of large loads. NERC should consider developing specific modeling and performance reliability standards for large loads, similar to MOD, PRC, and BAL standards applicable to large generators. This will ensure consistent treatment across balancing authorities, remove regulatory uncertainty for developers and transmission providers, and thus further the timely and

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29 ANOPR at P 31.

30 *Id*.

31 https://www.nerc.com/globalassets/programs/bpsa/alerts/2025/nerc-alert-level-2--large-loads.pdf.

orderly interconnection of large loads. The CAISO notes, however, that transmission providers like the CAISO may need to seek approval for their own reliability technical requirements in the meantime. Large loads are already proliferating, and some reforms may need to be implemented as soon as possible to address critical reliability needs, especially where transmission providers see local and regional needs.

III. CONCLUSION

The CAISO appreciates the Commission's efforts in establishing this proceeding, and respectfully requests that any final rule consider the CAISO's recommendations.

Respectfully submitted,

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CERTIFICATE OF SERVICE

I certify that I have served the foregoing document upon the parties listed on the official service list in the captioned proceedings, in accordance with the requirements of Rule 2010 of the Commission's Rules of Practice and Procedure (18 C.F.R. § 385.2010).

Dated at Folsom, California this 21st day of November, 2025.

/s/ Ariana Rebancos

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